that student was the brilliant mathematician Issai Schur, who kept the Frobenius tradition alive for decades to come.

In the meantime, Berlin was losing its reputation as the leading center of German mathematics to Göttingen. This did not endear Göttingen-style mathematics to Frobenius's heart, and he disaffectionately called it "a school, in which one amuses oneself more with rosy images than hard ideas" (quoted in [**11**, p. 47]). His aversion to the Göttingen patriarch Felix Klein and to Sophus Lie, according to one commentator, "knew no limits" (Biermann in [**11**, p. 47]).

Despite Frobenius's best efforts to keep the subject pure, representation theory has since been irretrievably polluted by applications in quantum physics and chemistry, crystallography, spectroscopy, and even virology. Worse still, thanks to the efforts of Hermann Weyl and Claude Chevalley, his beloved representation theory eventually merged with the theory of Lie groups. Frobenius thus was eventually reconciled with Lie, if only in the Platonic world of eternal mathematical objects.

4.12. Problems

Problem 4.12.1. Let G be the group of symmetries of a regular N-gon (it has $2N$ elements).

(a) Describe all irreducible complex representations of this group (consider the cases of odd and even N).

(b) Let V be the 2-dimensional complex representation of G obtained by complexification of the standard representation on the real plane (the plane of the polygon). Find the decomposition of $V \otimes V$ in a direct sum of irreducible representations.

Problem 4.12.2. Let p be a prime. Let G be the group of 3×3 matrices over $\mathbb{F}_p$ which are upper triangular and have 1's on the diagonal, under multiplication (its order is p^3). It is called the **Heisenberg group**. For any complex number z such that $z^p = 1$, we define a representation of G on the space V of complex functions on $\mathbb{F}_p$ by

$$(\rho \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} f)(x) = f(x-1),$$